

Adaptive Communication-Constrained DSGF for Efficient UAV Swarm Coordination

Abstract—Multi-UAV systems must cooperate under limited bandwidth, energy, and latency. Most multi-agent reinforcement learning pipelines optimize local policies while treating communication as a predefined radius graph or dense exchange, so messaging itself is not a first-class decision. We propose AC-DSGF, which treats the communication topology as a learnable decision variable and jointly optimizes task return with an explicit communication cost under a Top- K budget, atop residual spatial-temporal guidance. On 16-UAV cooperative navigation with five random seeds, AC-DSGF matches DSGF task performance while reducing communication mass by approximately $90\times$ and raising the Communication Efficiency Index. Budget sweeps, silence-collapse ablations, trigger analyses, and packet-loss tests indicate task-driven sparsity and graceful degradation, and demonstrate effectiveness under communication constraints.

Index Terms—Multi-UAV systems, multi-agent reinforcement learning, communication-efficient cooperation, dynamic graphs, residual guidance.

I. INTRODUCTION

a) Engineering reality.: In multi-UAV cooperation, every message costs bandwidth, energy, and latency. Controllers that optimize only local policies under an implicit “unlimited radio” assumption therefore leave a first-class resource unoptimized: **communication itself**.

b) Limitation of prior MARL guidance.: MAPPO-style CTDE stabilizes decentralized control [1], and graph guides (e.g., GAT) encode spatial relations for early coordination. A common pipeline is

policy $\rightarrow$ (predefined) communication $\rightarrow$ environment.

Topology is treated as given—by a radius, a dense graph, or a hand schedule—so agents cannot decide *whether* an exchange is currently needed.

c) Our shift.: We invert that dependency and learn topology as part of decision making:

environment $\rightarrow$ learn communication topology $\rightarrow$ policy.

Unlike previous approaches that assume communication topology is predefined, we treat topology selection as a learnable decision process.

Concretely, AC-DSGF answers: *when* to talk, *with whom*, and *how much* under a budget (Q1–Q3).

d) Method sketch.: **AC-DSGF** extends DSGF v2 with a soft gate g_{ij}^t , a Top- K budget layer $A_t^{\text{AC}} = A \odot E^K$, and the joint objective $\mathbb{E}[\sum r_t] - \lambda_c \mathbb{E}[\sum C_t]$, while residual guidance $a = \pi(o) + \beta \Delta(\Phi)$ keeps communication assistive rather than dominant (Fig. 1).

e) Claim.: We do not position AC-DSGF as a Success maximizer. Experiments show it **achieves comparable task performance** to DSGF while **reducing communication overhead by approximately $90\times$** (Table V).

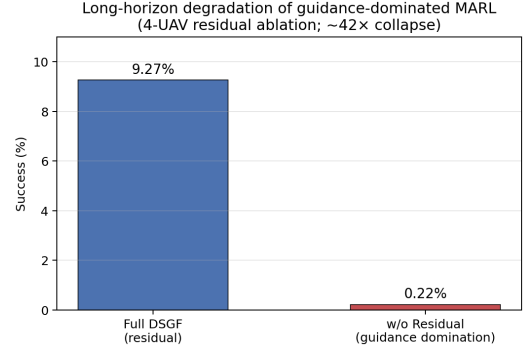


Fig. 1. Motivation: without residual decoupling, guided MARL can collapse under long-horizon optimization (4-UAV ablation; Table VI). AC-DSGF keeps residual guidance while adding budget-aware communication.

f) Contributions.:

- 1) **Learnable communication topology**: $g_{ij}^t = \sigma(W[h_i, h_j, d_{ij}, \rho_{ij}])$ on radius support, so *task need*—not distance alone—decides who talks.
- 2) **Joint optimization under a communication budget**: $J = \mathbb{E}[\sum r] - \lambda_c \mathbb{E}[\sum C]$ with Top- K adjacency $A_t^{\text{AC}} = A \odot E_t^K$, co-optimizing control and messaging rather than pruning after policy learning.
- 3) **Residual-guided stable sparse communication**: $a = \pi(o) + \beta \Delta(\Phi)$ keeps guidance assistive, mitigating silence collapse and topology degeneration under tight budgets.

Evaluation at $N=16$ (five seeds) covers comparison, budget Pareto, silence-collapse, trigger/stability, compute cost, and packet-loss robustness.

II. RELATED WORK

We structure prior work around three gaps. The unifying distinction is: *previous methods largely treat communication as given; AC-DSGF treats communication as learned*.

A. Multi-Agent RL for UAV Swarms

Centralized Training with Decentralized Execution (CTDE) methods such as MAPPO, MADDPG, and value-factorization approaches (e.g., QMIX) stabilize cooperative control by sharing critics or mixing networks in training while keeping actors local at execution [1], [2], [3]. Aerial applications demonstrate navigation and collision handling under partial observability [4], [5]. These pipelines, however, typically optimize task return under an implicit assumption of **full or sufficient communication**: either agents act independently given a shared training signal, or coordination depends on

centralized features that do not survive bandwidth, energy, and latency limits of large UAV swarms.

Gap 1. Existing MARL methods for UAV cooperation mainly optimize policy performance while overlooking communication constraints in large-scale swarms.

B. Graph-based Communication and Interaction Modeling

Graph neural networks and attention inject relational inductive bias into MARL [6], [7], [8], [9]. GAT-style guides and graph Transformers adaptively weight neighbors and capture time-varying spatial relations. Critically, **attention is not a communication decision**: it aggregates features under a *predefined* support—a radius mask, a dense graph, or a hand schedule. Dense message passing costs $\mathcal{O}(N^2)$; binary radius graphs still exchange on *all* in-range edges regardless of task urgency. Our prior DSGF improves quality weighting and residual guidance, yet still activates the full geometric support whenever $A_{ij}=1$.

Gap 2. Existing graph attention mechanisms mainly aggregate information assuming predefined communication structures; they do not equate topology adaptation with bandwidth-aware economy.

C. Communication-Efficient MARL

A parallel literature studies message pruning, gated talk/silence decisions, dropout, and bandwidth-aware objectives [10], [11], [12], [13], [14], [15], [16]. Many approaches rely on **stochastic deletion** $P(\text{drop})$ or heuristic schedules weakly coupled to continuous control. Random pruning can discard coordination-critical links under tight budgets; over-aggressive silence collapses information flow. In contrast, AC-DSGF learns a continuous gate $g_{ij}^t = \sigma(W[h_i, h_j, d_{ij}, \rho_{ij}])$ jointly with the policy under an explicit cost $C = \sum g_{ij}$ and optional Top- K budgeting—sparsity is *task-driven*, not random dropout.

Gap 3. Prior communication-efficient MARL often **drops messages** rather than *learning when and with whom* communication is necessary for the ongoing task under a quantified budget.

a) *Positioning.*:

Previous Communication is *given* (radius / dense / random drop)

Ours Communication is *learned* (task-aware gate + budget)

At discrete time t , agent $i \in \{1, \dots, N\}$ receives a local observation

$$o_i^t = [p_i^t, v_i^t, g_i^t, \ell_i^t] \in \mathbb{R}^{d_o}, \quad (1)$$

where $p_i^t, v_i^t \in \mathbb{R}^2$ are position and velocity, g_i^t is the goal-relative vector, and ℓ_i^t denotes onboard range sensing. Each agent executes a continuous action $a_i^t \in \mathcal{A}$ under a decentralized policy with centralized training (CTDE).

The team seeks to maximize expected cumulative reward under communication cost,

$$\max_{\theta} \mathbb{E} \left[\sum_{t=0}^H r_t \right] - \lambda_c \mathbb{E} \left[\sum_{t=0}^H C_t \right], \quad r_t = \sum_{i=1}^N r_i^t, \quad (2)$$

where $r_i^t = r_i^{\text{goal}}(t) + r_i^{\text{coll}}(t)$ and C_t is defined below. When $\lambda_c=0$, Eq. (2) reduces to standard return maximization (baselines). For AC-DSGF, $\lambda_c > 0$ makes communication a first-class training signal.

B. Communication Constraint

Airborne links are range-limited and capacity-constrained. Two agents may exchange messages only if $\|p_i^t - p_j^t\| < R_c$, inducing a feasible mask $A^t \in \{0, 1\}^{N \times N}$. We further impose a soft communication budget through a learnable gate $g_{ij}^t \in [0, 1]$. Unless stated otherwise, the communication cost used in tables is the *time-averaged soft gate mass*

$$C_t = \sum_{i \neq j} g_{ij}^t, \quad C = C_{\text{comm}} = \frac{1}{T} \sum_{t=1}^T C_t = \frac{1}{T} \sum_{t=1}^T \sum_{i \neq j} g_{ij}^t. \quad (3)$$

For un-gated baselines (GAT/DSGF), we report the analogous average number of active geometric edges $\frac{1}{T} \sum_t \sum_{i,j} A_{ij}^t$ (or quality-weighted support) so that Comm columns are comparable as *message intensity*, not raw radio packet counts. The Table I AC-DSGF value $C \approx 0.43$ is therefore the mean soft mass under Eq. (3) over the 200-ep evaluation—not an undefined unit. Eq. (2) pairs this C with task return via λ_c . Hard budgets additionally restrict support via $A_t^{\text{AC}} = A_t^t \odot E_t^K$ (Sec. IV-B).

AC-DSGF jointly addresses (i) task performance, (ii) learned topology via g_{ij}^t , and (iii) explicit budgets $C = \sum g_{ij}$ with Top- K , while residual DSGF guidance keeps sparse messaging assistive (Secs. VI-D–VI-E). It is therefore not “DSGF + ℓ_2 pruning,” but a communication-constrained coordination stack.

III. PROBLEM FORMULATION

A. Multi-UAV Cooperative Navigation

Consider a team of N UAVs performing cooperative navigation in a continuous 2D workspace under partial observability.

To compare methods under different messaging intensities, we report

$$\text{CEI} = \frac{S}{C_{\text{comm}} + \varepsilon}, \quad \varepsilon = 10^{-6}, \quad (4)$$

where S is the evaluation success rate. MAPPO has no explicit graph messaging ($C_{\text{comm}}=0$) and CEI is marked as not applicable. Our primary claim is *not* that AC-DSGF maximizes S unconditionally, but that it preserves **comparable** S while substantially reducing C_{comm} relative to fixed-topology guidance.

TABLE I
SUMMARY OF KEY SYMBOLS. SOFT GATES $g_{ij}^t \in [0, 1]$ ARE CONTINUOUS (PROBABILITY-LIKE INTENSITIES), NOT HARD BINARY EDGES; BINARY DECISIONS APPEAR ONLY AFTER TOP- K WHEN A HARD BUDGET IS ENFORCED.

Symbol	Meaning
N	Number of UAVs (agents)
o_i^t	Local observation of agent i at time t
h_i^t	Latent embedding of o_i^t
A^t / A_{ij}^t	Feasible radius topology (binary mask)
q_{ij}^t	Communication quality score
g_{ij}^t	Soft communication gate in $[0, 1]$ (not binary)
K / K_i	(Per-agent) communication budget / Top- K width
A_t^{AC}	Budgeted adjacency $A^t \odot E_t^K$
e_{ij}^t	Effective edge weight for attention
$G_t = (V, E_t)$	Communication graph at time t
α_{ij}^t	Attention weight on edge (i, j)
C_t / C	Instantaneous / time-averaged soft communication cost
λ_c	Communication cost weight in the joint objective
Φ_i^t	Residual guidance feature
$\beta(t)$	Residual mixing schedule
a_i^t	Action: $\pi(o_i) + \beta\Delta(\Phi_i)$

NOTATION

IV. PROPOSED METHOD: AC-DSGF

We propose *Adaptive Communication-aware Dynamic Sparse Graph Fusion* (AC-DSGF). Unlike post-hoc message pruning, AC-DSGF **jointly** learns the control policy and a task-dependent communication topology under an explicit cost. Fig. 2 shows three layers: (i) observation embedding, (ii) adaptive communication (gate + budget), (iii) residual DSGF policy.

A. Joint Communication–Control Objective

Let $r_t = \sum_i r_i^t$ denote the team reward at time t and

$$C_t = \sum_{i \neq j} g_{ij}^t \quad (5)$$

the soft communication mass. AC-DSGF optimizes

$$\max_{\theta} \mathbb{E} \left[\sum_{t=0}^H r_t \right] - \lambda_c \mathbb{E} \left[\sum_{t=0}^H C_t \right], \quad (6)$$

i.e., communication is optimized *together with* policy learning. This differs from training a dense communicator and then pruning messages at deployment time.

B. Adaptive Communication Graph

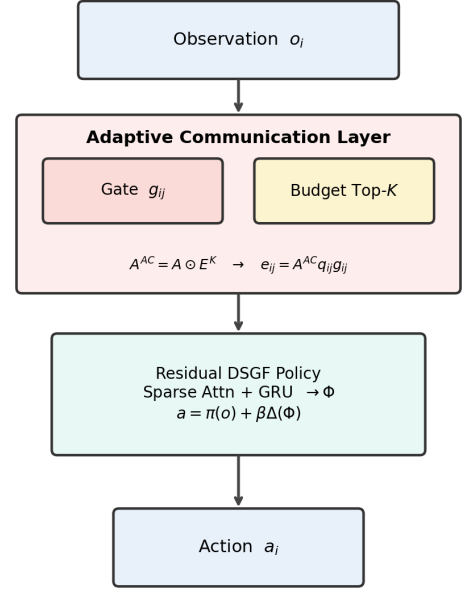
a) *Feasible mask and quality.*: The radius mask $A_{ij}^t = \mathbb{I}[\|p_i^t - p_j^t\| < R_c]$ and quality $q_{ij}^t \in (0, 1)$ yield support $\tilde{A}_{ij}^t = A_{ij}^t q_{ij}^t$.

b) *Gate (learned topology).*: Let $h_i^t = \psi(o_i^t)$ and let ρ_{ij}^t encode an interaction-risk feature (relative closing speed / proximity prior; implemented via the edge-state channels alongside d_{ij}^t and q_{ij}^t). The communication gate is

$$g_{ij}^t = \sigma \left(W_g [h_i^t; h_j^t; d_{ij}^t; \rho_{ij}^t] \right) \cdot A_{ij}^t \in [0, 1], \quad (7)$$

where W_g denotes the parameters of a lightweight MLP (bias omitted). Unlike random dropout, g_{ij}^t is a *learned* function of

AC-DSGF Framework (three layers)



$$\text{Joint objective: } \max \mathbb{E}[\sum r_t] - \lambda_c \mathbb{E}[\sum C_t]$$

Fig. 2. Framework of AC-DSGF: adaptive communication (gate + budget) defines the topology consumed by residual DSGF guidance, rather than post-hoc pruning.

agent states and risk—empirically aligned with proximity risk (Sec. VI-E).

c) *Hard budget $\Rightarrow$ AC adjacency.*: Given budget ratio $\rho \in (0, 1]$,

$$K_i = \lceil \rho |\mathcal{N}_i| \rceil, \quad E_t^K = \text{TopK}_{K_i}(\{g_{ij}^t\}_{j \in \mathcal{N}_i}), \quad (8)$$

and the adaptive communication adjacency

$$A_t^{AC} = A^t \odot E_t^K, \quad e_{ij}^t = A_{ij}^{AC} q_{ij}^t g_{ij}^t. \quad (9)$$

Training may use soft g with the cost in Eq. (6); evaluation can enforce hard Top- K without retraining (budget sweep).

d) *Difference from pruning.*: Message pruning typically removes links after (or independently of) policy learning. AC-DSGF changes the *information topology* inside the forward pass: setting $g=A$ (“no-budget”) restores near-full local broadcast and inflates C_t by over three orders of magnitude with only a small Success change (Sec. VI-D). Hence AC-DSGF is not “DSGF + ℓ_2 regularization,” but a gated topology learner under a joint objective.

C. Residual Spatial–Temporal Guidance

Sparse attention aggregates neighbors under e^t :

$$\alpha_{ij}^t \propto \exp(f(h_i^t, h_j^t)) e_{ij}^t, \quad h_i^{t'} = \sigma \left(\sum_j \alpha_{ij}^t W h_j^t \right), \quad (10)$$

followed by $z_i^t = \text{GRU}(h_i^{t'}, z_i^{t-1})$ and guidance Φ_i^t . Actions use residual decoupling

$$a_i^t \sim \pi_{\theta}(\cdot | o_i^t) + \beta(t) \Delta(\Phi_i^t), \quad (11)$$

TABLE II
HOW COMMUNICATION IS TREATED DURING LEARNING.

Method	Communication mechanism
Dropout / random pruning	Stochastic deletion of messages
Post-hoc / penalty-only	Constraint applied after (or around) a fixed policy graph
GAT / radius graphs	Attention on a <i>fixed</i> geometric adjacency
AC-DSGF	Task-driven gate + budget; topology is co-learned

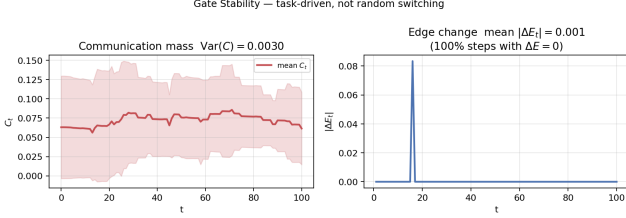


Fig. 3. Gate stability: mean $\pm$ std communication mass C_t and mean $|\Delta E_t|$ over episodes. Sparse AC graphs are dynamic yet non-chattering.

so coordination *assists* rather than dominates execution (Table VI: 9.27% $\rightarrow$ 0.22% without residual).

D. Difference from Conventional Communication Learning

Table II contrasts AC-DSGF with common alternatives. Key distinction: the communication topology is optimized as a **decision variable** rather than treated as a fixed network structure. Joint optimization (Eq. (6)) and the explicit map $A \rightarrow g \rightarrow A^{\text{AC}} \rightarrow e$ make gating part of the forward computation, not an ℓ_2 -style afterthought on an otherwise dense GAT/DSGF stack.

E. Gate Stability (non-random switching)

A potential concern is that sparse gates might chatter and destabilize training. Using frozen AC-DSGF rollouts, we measure soft cost variance $\text{Var}(C_t)$ and the edge-change rate

$$\Delta E_t = ||E_t| - |E_{t-1}||. \quad (12)$$

Empirically (Fig. 3; 12 episodes), $\text{Var}(C_t) \approx 3 \times 10^{-3}$ and $>99\%$ of steps have $\Delta E_t = 0$: C_t varies smoothly with the task phase while edges rarely chatter. Communication adapts when proximity risk changes (Sec. VI-E), rather than randomly toggling.

F. Theoretical Remarks

We do not claim global optimality of the joint objective. Instead we record two structural properties that explain the empirical efficiency–performance trade-off.

a) *Lemma 1 (Communication sparsification)*.: Under a per-agent Top- K budget with average degree K , the attended communication support has size $\mathcal{O}(NK)$. Hence messaging complexity reduces from dense pairwise $\mathcal{O}(N^2)$ (or full in-radius $\mathcal{O}(Nk)$ with $k \rightarrow N$) to $\mathcal{O}(NK)$ when K is held approximately constant. A full expansion appears in Sec. IV-H.

Algorithm 1 AC-DSGF Training

Require: Observations $\{o_i^t\}$, radius R_c , budget ratio ρ (or K), weight λ_c

Require: Policy parameters θ , gate parameters ϕ (collectively θ, ϕ in text)

Ensure: Updated θ, ϕ

for each training iteration **do**

- 2: Collect rollouts with decentralized actors
- 3: **for** each timestep t in batch **do**
- 4: Embed $h_i^t \leftarrow \psi(o_i^t)$; build candidate mask A^t and quality q^t
- 5: **Gate:** $g_{ij}^t \leftarrow \sigma(W_g[h_i; h_j; d_{ij}; \rho_{ij}]) \cdot A_{ij}^t$
- 6: **Budget:** $E_t^K \leftarrow \text{TopK}_{K_i}(\{g_{ij}^t\})$; $A_t^{\text{AC}} \leftarrow A^t \odot E_t^K$
- 7: Build $e_{ij}^t \leftarrow A_{ij}^{\text{AC}} q_{ij}^t g_{ij}^t$; $C_t \leftarrow \sum_{ij} g_{ij}^t$
- 8: Aggregate neighbors with sparse attention on e^t ; $\text{GRU} \rightarrow \Phi_i^t$
- 9: **Policy:** sample $a_i^t \sim \pi_\theta(\cdot | o_i^t) + \beta(t)\Delta(\Phi_i^t)$
- 10: **end for**
- 11: Update θ, ϕ by PPO/GAE loss plus $\lambda_c \mathbb{E}[\sum_t C_t]$
- 12: **end for**

b) *Lemma 2 (Residual stability)*.: If guidance enters as an additive residual $a = \pi(o) + \beta\Delta(\Phi)$ with scheduled $\beta \in [0, 1]$, the local policy remains a well-defined fallback when the gated graph shrinks ($g \rightarrow 0$ or $K \rightarrow 1$). Thus budgeted topology control cannot erase the base controller—empirically preventing the degeneration observed when residual decoupling is removed (Table VI).

c) *Performance under budget (informal)*.: Let π^* be an unrestricted dense communicator and π_{AC} the AC-DSGF policy under budget B on $\mathbb{E}[\sum C_t]$. Joint training targets $J(\pi_{\text{AC}}) \geq J(\pi^*) - \varepsilon(B)$ for a budget-dependent gap ε , rather than unconstrained Success maximization. Budget sweeps (Sec. VI-C) show graceful degradation of Success as B decreases, consistent with this view.

G. Optimization

Actors use clipped PPO + GAE with the AC cost term for $\lambda_c = 10^{-3}$ (frozen). Algorithm 1 makes the Gate $\rightarrow$ Budget $\rightarrow$ Policy order explicit. Computational analysis is given in Sec. IV-H.

H. Computational and Communication Complexity

Let N be the number of UAVs, d the hidden width, $\mathcal{N}_i$ the feasible neighbors of agent i under radius R_c , and $k = \frac{1}{N} \sum_i |\mathcal{N}_i|$ the average degree ($k \ll N$ in sparse swarms). Let $K_i = \lceil \rho |\mathcal{N}_i| \rceil$ denote the Top- K budgeted degree and $K = \frac{1}{N} \sum_i K_i$ the average budgeted degree ($K \leq k$).

a) *Dense / GAT baselines*.: Full pairwise messaging or dense attention costs $\mathcal{O}(N^2 d)$ per layer. Even with a binary radius mask, GAT-style aggregation that treats all in-radius neighbors equally still processes $\Theta(Nk)$ edges; under moderate R_c this rapidly approaches quadratic growth in N .

TABLE III
COMPLEXITY AND TYPICAL COMMUNICATION INTENSITY
(CONCEPTUAL). K IS THE AVERAGE BUDGETED DEGREE; k IS THE
AVERAGE RADIUS DEGREE.

Method	Communication	Aggregation	Complexity
MAPPO	none	—	$\mathcal{O}(Nd)$
GAT / Transformer	high	dense / full	$\mathcal{O}(N^2d)$
DSGF	medium (all radius links)	sparse k	$\mathcal{O}(Nkd + Nd)$
AC-DSGF	low (Top- K gates)	sparse $K \leq k$	$\mathcal{O}(Nkd + Nd)$

TABLE IV
RUNTIME / SCALABILITY (CPU, NO RETRAINING). INFERENCE MS/STEP
AT $N=16$ (50 TIMED STEPS); COMM FROM TABLE V. FOR $N=4, 8$:
ATTENDED COMPLEXITY ONLY ($K \ll k$).

N	GAT	DSGF	AC-DSGF
4	$\mathcal{O}(Nk)$	$\mathcal{O}(Nk)$	$\mathcal{O}(NK)$
8	$\mathcal{O}(Nk)$	$\mathcal{O}(Nk)$	$\mathcal{O}(NK)$
16	1.66 ms / 38.8	3.35 ms / 39.3	3.94 ms / 0.43

b) **AC-DSGF**: Gate evaluation runs on the feasible edge set: $\mathcal{O}(|E|d) = \mathcal{O}(Nkd)$. Top- K budgeting per agent costs $\mathcal{O}(k \log k)$ (or $\mathcal{O}(K \log k)$ with partial selection), totaling $\mathcal{O}(Nk \log k)$. Sparse attention then operates on the budgeted support of size $\mathcal{O}(NK)$:

$$\mathcal{O}(Nkd + Nd) \quad (\text{attention} + \text{GRU}), \quad (13)$$

which approaches linear scaling in N when K is held approximately constant by the budget ratio. Because $K \ll N$ under tight budgets, the relative advantage versus $\mathcal{O}(N^2d)$ dense attention **widens as team size grows**. Communication volume itself is bounded by $\sum_i K_i = \mathcal{O}(NK)$, whereas GAT/DSGF without gating use $\mathcal{O}(Nk)$ open geometric edges. Pairwise distance masking remains $\mathcal{O}(N^2)$ and is shared by all geometric graph methods; the practical gain of AC-DSGF is in *attended* edges and transmitted soft mass under $K \ll N$. Fig. 6 illustrates communication versus team size. Table IV reports CPU inference at $N=16$ alongside Table V Comm: the gate adds modest latency while cutting soft mass by nearly two orders of magnitude versus radius-support DSGF/GAT.

V. EVALUATION UNDER COMMUNICATION-CONSTRAINED SWARM COORDINATION

We evaluate whether AC-DSGF can **maintain comparable task performance** while **learning a task-aware sparse communication topology**. Success alone is not the primary score; we jointly report Success, Comm, and CEI.

A. Environment and Protocol

All formal results use the VMAS continuous multi-agent navigation scenario. Unless stated otherwise:

- Training: 102,400 frames; evaluation: 200 episodes.
- Main table seeds: {42, 3407, 2026, 1234, 8888} ($N=16$).
- Horizon $H=128$; success threshold 0.3 world units; $R_c=0.5$.
- DSGF v2 backbone and AC hyperparameters are frozen (dsgf_v2_frozen.yaml, ac_dsgf_16uav.yaml).

TABLE V
TASK PERFORMANCE UNDER COMMUNICATION CONSTRAINTS ($N=16$, 5
SEEDS, 200-EP EVAL). AC-DSGF MAINTAINS COMPARABLE TASK
COMPLETION WITH SUBSTANTIALLY REDUCED COMMUNICATION
OVERHEAD (HIGHER CEI). MEAN $\pm$ STD.

Method	Success (%)	Collision (%)	Comm	CEI
MAPPO	2.40 $\pm$ 0.90	32.3 $\pm$ 1.9	0.00 $\pm$ 0.00	—
GAT	2.04 $\pm$ 0.90	29.6 $\pm$ 1.0	38.80 $\pm$ 1.99	0.0005
Transformer	0.70 $\pm$ 0.16	30.8 $\pm$ 3.0	240.00 $\pm$ 0.00	≈ 0
DSGF	4.01 $\pm$ 1.68	31.5 $\pm$ 3.3	39.27 $\pm$ 1.92	0.0010
AC-DSGF	3.95 $\pm$ 0.83	30.8 $\pm$ 1.4	0.43 $\pm$ 0.09	0.0913

TABLE VI
DSGF RESIDUAL ABLATION AT $N=4$ (MECHANISM TEST; *not* TABLE I
SCALE).

Variant	Success (%)
w/o Dynamic Graph	0.28
w/o Temporal	0.86
w/o Residual	0.22
Full DSGF	9.27

B. Baselines

MAPPO (no graph messaging); GAT-MAPPO (binary radius graph + alignment reward); Transformer / full attention; DSGF v2 (quality-weighted sparse graph + residual); **AC-DSGF** (DSGF v2 + learned gate + $\lambda_c C_{\text{comm}}$).

C. Metrics

Success S , collision rate, episodic reward, communication cost C_{comm} (Eq. (3)), and $\text{CEI} = S/(C_{\text{comm}} + \varepsilon)$ (Eq. (4)). Demo videos are qualitative and *not* used as Table metrics.

D. Experiment Map

- 1) **Overall performance** (Table V): Success / Comm / CEI at $N=16$.
- 2) **Communication budget** (Table VII, Fig. 4, Fig. 8): Pareto under constrained messaging.
- 3) **Ablation** (Table VI, Table VIII): residual, budget, and learned vs. fixed/random topology.
- 4) **Communication behavior** (Fig. 5, Fig. 3): risk-driven sparsity, non-chattering gates.
- 5) **Scalability & robustness** (Fig. 6, Fig. 7): $\mathcal{O}(NK)$ story and packet loss.

VI. RESULTS

A. Overall Performance

Reading Table V. We do *not* claim that AC-DSGF outperforms DSGF on Success. Instead, AC-DSGF **maintains comparable task performance** (3.95% vs. 4.01%) while reducing communication cost by approximately **two orders of magnitude** ($39.27 \rightarrow 0.43$, $\sim 91\times$) and lifting CEI by nearly $90\times$ (Fig. 8). The primary axes are therefore Comm and CEI under a Success-hold constraint.

TABLE VII

SUCCESS (%) UNDER COMMUNICATION BUDGET RATIO ρ ($N=16$, SEED 42, 64-EP DIAGNOSTIC EVAL; QUALITATIVE CONSISTENCY WITH TABLE I CEI RANKING). SOFT-COST SCALES DIFFER FROM TABLE I; WE REPORT TRENDS, NOT ABSOLUTE % INTO THE MAIN CLAIM.

Budget ρ	GAT	DSGF	AC-DSGF
100%	21.3	21.7	27.1
50%	18.2	21.6	25.2
10%	16.9	21.7	24.2

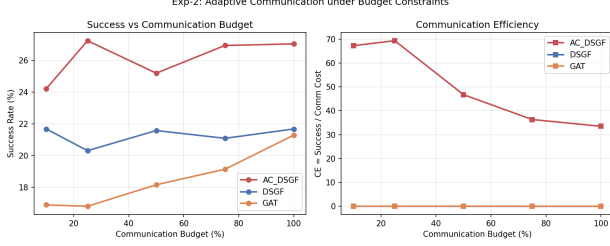


Fig. 4. Success vs. communication budget ($N=16$). AC-DSGF maintains task effectiveness under severe budget cuts (graceful degradation).

B. Ablation Study

This ablation is not merely “remove a module $\Rightarrow$ score drops.” It isolates a failure mode of guided MARL: when Φ dominates the actor, policies overfit coordination surrogates and lose long-horizon task competence. Residual decoupling keeps guidance *assistive*, explaining the $\sim 42\times$ Success collapse without it. (AC gate ablations appear in Sec. VI-D.)

C. Communication Budget Analysis

Under a 90% budget cut (100% $\rightarrow$ 10%), AC-DSGF success drops only $\sim 11\%$ relatively (27.1 $\rightarrow$ 24.2), i.e., *graceful degradation*. Paper wording prefers “over three orders of magnitude” for gate-level soft-cost reductions versus “no-budget” ablation, rather than advertising un-normalized $5000\times$ factors without defining soft cost.

D. Silence Collapse Ablation

Interpretation. Sparsity is *not* silence collapse. AC-full trades a few percentage points of Success for over-three-orders reduction in Comm relative to full local broadcast—matching the locked claim of comparable performance at far lower messaging.

E. Communication Behavior Analysis

Pearson correlations with Comm: goal distance 0.03, dispersion -0.57 , nearest-neighbor risk **0.84**. When interaction risk is high, communication intensifies; when the swarm disperses, messaging attenuates. Together with gate stability (Fig. 3; $>99\%$ steps with $\Delta E_t=0$), this shows that sparsity is *structured* rather than random pruning or uncontrolled silence.

F. Scalability Analysis

Fig. 6 places the Table I communication result of AC-DSGF in a scaling context: maintaining task effectiveness does not require paying the dense-graph communication tax as N grows.

TABLE VIII

COMMUNICATION MECHANISM VALIDATION ($N=16$, AC CHECKPOINT). AC-FULL IS *not* SILENCE COLLAPSE: OPENING ALL RADIUS EDGES ($g=A$) INFLATES COMM BY OVER THREE ORDERS OF MAGNITUDE WITH ONLY A SMALL SUCCESS GAIN.

Variant	Success (%)	Comm	CEI
AC-full (learned)	24.6	0.008	32.0
AC-no budget ($g=A$)	28.9	40.6	0.007
AC-random gates	28.5	20.3	0.014

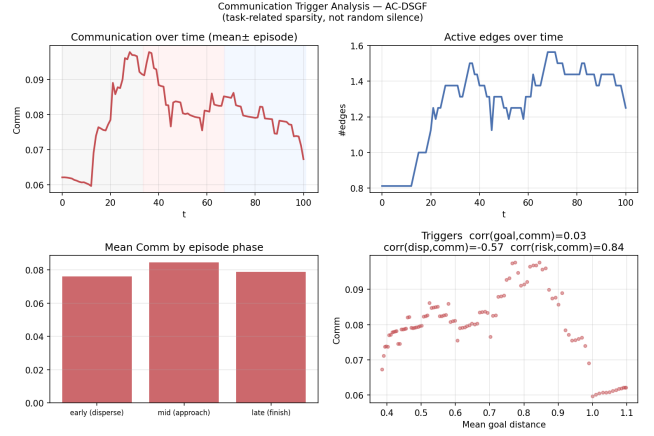


Fig. 5. Communication behavior: Comm rises with proximity risk ($\text{corr}=0.84$) and falls with spatial dispersion ($\text{corr}=-0.57$)—sparse topology is task-driven, not random.

G. Robustness under Packet Loss

From $p=0$ to $p=0.7$, GAT Success drops from 21.2% to 15.8% (relative $\sim 25\%$), while AC-DSGF moves only from 26.2% to 25.8%. MAPPO is insensitive by construction (no graph messaging). We interpret AC-DSGF’s flat curve as **graceful degradation**: messaging was already sparse and task-conditioned, so random drops remove fewer coordination-critical edges on average.

H. Communication–Performance Pareto

LIMITATIONS

Absolute Success at $N=16$ remains low for all methods; claims are comparative and communication-centric. Budget/silence diagnostics may use different episode counts than Table I and must be described as such. Reported $C = \sum g_{ij}$ is a soft intensity proxy, not a physical packet counter; mapping to radio frames is left to hardware deployment, while packet-loss diagnostics already probe robustness to random edge drops.

VII. CONCLUSION

We presented AC-DSGF for communication-constrained multi-UAV cooperative navigation. By jointly optimizing task return and communication mass $\mathbb{E}[\sum r_t] - \lambda_c \mathbb{E}[\sum C_t]$, learning a task-dependent gate, and enforcing Top- K budgets atop residual DSGF guidance, AC-DSGF **maintains task effectiveness while reducing unnecessary communication**.

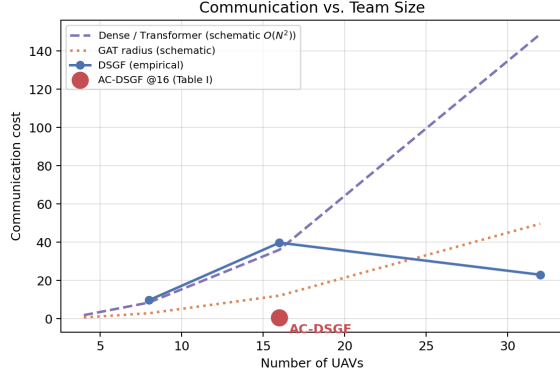


Fig. 6. Communication versus number of UAVs. Dense / GAT schemas grow quickly; DSGF uses geometric support; AC-DSGF at $N=16$ sits far lower (Table V), consistent with $\mathcal{O}(NK)$ budgeted messaging (Sec. IV-H).

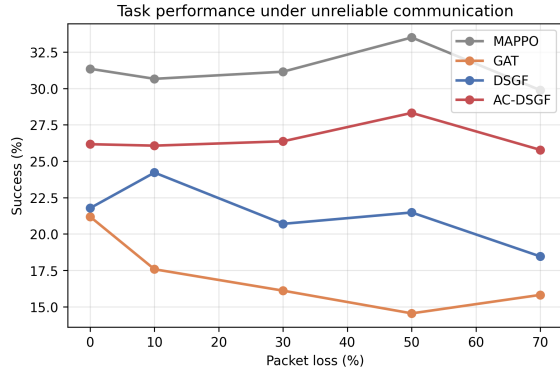


Fig. 7. Task success under random edge drops ($N=16$, 64-ep diagnostic). AC-DSGF remains nearly flat as loss increases, whereas GAT degrades more noticeably—consistent with *graceful degradation* under unreliable links.

At $N=16$, it achieves Success comparable to DSGF with approximately $90\times$ lower communication cost, with supporting budget, silence-collapse, trigger, compute-cost, and packet-loss analyses. Future work includes ROS / Gazebo hardware-in-the-loop validation and communication-aware motion planning, where agents may actively approach teammates to obtain high-value information.

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TABLE IX
SUCCESS (%) VS. PACKET-LOSS RATE p ($N=16$, SEED 42, 64 EPS). WE EMPHASIZE TREND (DEGRADATION SLOPE), NOT ABSOLUTE COMPARISON WITH TABLE I.

Method	0%	10%	30%	50%	70%
MAPPO	31.4	30.7	31.2	33.5	29.9
GAT	21.2	17.6	16.1	14.6	15.8
DSGF	21.8	24.2	20.7	21.5	18.5
AC-DSGF	26.2	26.1	26.4	28.3	25.8

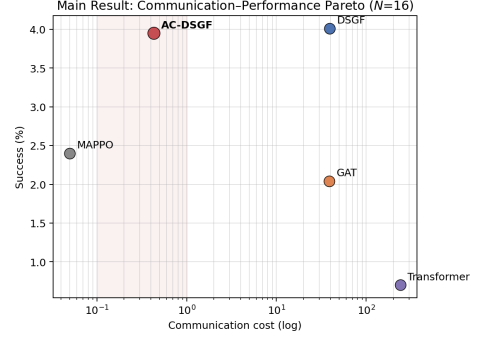


Fig. 8. Communication-performance Pareto ($N=16$, Table V). AC-DSGF occupies the high-efficiency region (left): comparable Success at minimal communication cost.

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